

PBA: 2.

Data Visualization

Fall 2026

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Where are we? Where are we going?

- **Where we have been**

- Week 1: R and RStudio, functions and packages, Quarto
- Assigned readings for Weeks 1–2: the R basics (operators, types, vectors, data frames); we start using them today

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- **Today: a first look at data visualization**
 - This is how R is learned here: on data, from the first week. The grammar comes when you need it; nothing today assumes prior programming
 - Slides and the R script are on the course website every week; after the lecture you run today's code yourself, then E1 at your own pace with the TAs

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- **Where we are going**

- Next week: data wrangling. Then causality, prediction, regression, and uncertainty

Visualization → Wrangling → Causality → Prediction → Regression → Uncertainty

Roadmap

1. **A first look at data, and a first plot**

- Midwest counties: one row is a county

2. **Anatomy of a ggplot**

- Building plots by layers
- Geoms (i.e., geometric objects) are functions!
- Adding informative labels
- Mapping versus setting aesthetics

3. **Histograms/boxplots and grouped data**

- Histogram and boxplots
- Grouped data

Rest of the class: In-class exercise E1 (Gapminder)

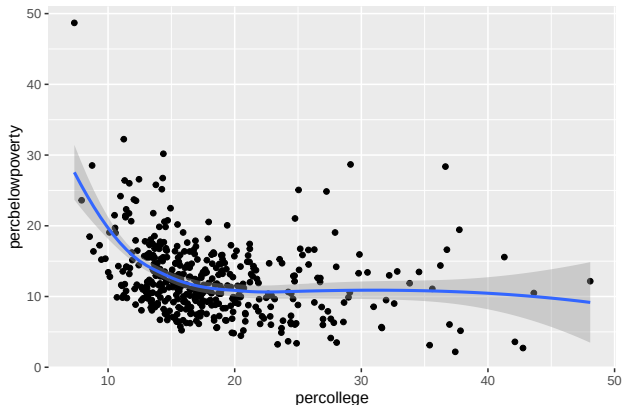
Midwest Data

```
1 # Install once: install.packages(c("pacman", "ggplot2"))
2 # Load with library() or require(), or let pacman do both:
3 > pacman::p_load(ggplot2)
4 > midwest
5 # A tibble: 437 × 28
6   PID county state area poptotal popdensity popwhite popblack popamerindian popasian popother
7   <int> <chr> <chr> <dbl> <int> <dbl> <int> <int> <int> <int> <int>
8 1 561 ADAMS IL 0.052 66090 1271. 63917 1702 98 249 124
9 2 562 ALEXAN IL 0.014 10626 759 7054 3496 19 48 9
10 3 563 BOND IL 0.022 14991 681. 14477 429 35 16 34
11 4 564 BOONE IL 0.017 30806 1812. 29344 127 46 150 1139
12 5 565 BROWN IL 0.018 5836 324. 5264 547 14 5 6
13 # i 434 more rows
14 # i 17 more variables: percwhite <dbl>, percblack <dbl>, percamerindian <dbl>, percasian <dbl>, ...
```

- The table comes with ggplot2; nothing to type in.
- One row is a **county**: 437 counties in five Midwest states (IL, IN, MI, OH, WI).
- Two columns for the first plots: percollege (% with a college degree), percbelowpoverty (% below the poverty line).
- <int>: integer; <chr>: text; <dbl>: numeric.

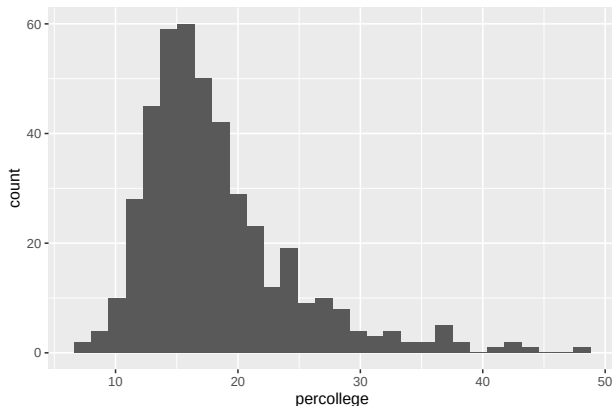
College Education and Poverty

```
1 > ggplot(midwest, mapping = aes(x = percollege, y = percbelowpoverty)) +  
2   geom_point() + geom_smooth() # the curve: loess, local smoothing
```



A Histogram of College Education

```
1 > ggplot(midwest, mapping = aes(x = percollege)) +  
2   geom_histogram()
```



Two lines of code drew that.

What did each piece do?

The rest of today: the anatomy of a ggplot, the pieces you will use every week, then time to build your own.

1/ Anatomy of a ggplot

Midwest Data

```
1 > ggplot2::midwest ## If needed: install.packages("ggplot2")
2
3 # A tibble: 437 × 28
4   PID county state area poptotal popdensity popwhite popblack popamerindian popasian popother
5   <int> <chr> <chr> <dbl> <int> <dbl> <int> <int> <int> <int> <int>
6 1 561 ADAMS IL 0.052 66090 1271. 63917 1702 98 249 124
7 2 562 ALEXAN IL 0.014 10626 759 7054 3496 19 48 9
8 3 563 BOND IL 0.022 14991 681. 14477 429 35 16 34
9 4 564 BOONE IL 0.017 30806 1812. 29344 127 46 150 1139
10 5 565 BROWN IL 0.018 5836 324. 5264 547 14 5 6
11 6 566 BUREAU IL 0.05 35688 714. 35157 50 65 195 221
12 7 567 CALHOUN IL 0.017 5322 313. 5298 1 8 15 0
13 8 568 CARROLL IL 0.027 16805 622. 16519 111 30 61 84
14 9 569 CASS IL 0.024 13437 560. 13384 16 8 23 6
15 10 570 CHAMPA IL 0.058 173025 2983. 146506 16559 331 8033 1596
16 # i 427 more rows
17 # i 17 more variables: percwhite <dbl>, percblack <dbl>, percamerindian <dbl>, percasian <dbl>,
18 # percother <dbl>, popadults <int>, perchsds <dbl>, percollege <dbl>, percprof <dbl>,
19 # poppovertyknown <int>, percpovertyknown <dbl>, percbelowpoverty <dbl>,
20 # percchildbelowpovert <dbl>, percadultpoverty <dbl>, percelderlypoverty <dbl>, inmetro <int>,
21 # category <chr>
22 # i Use `print(n = ...)` to see more rows
```

Building Up a Graph in Pieces

- Create a ggplot object and direct it to the correct data:

```
1 > p <- ggplot(data = midwest)
```

- **Mapping:** tell ggplot what visual aesthetics correspond to which variables:

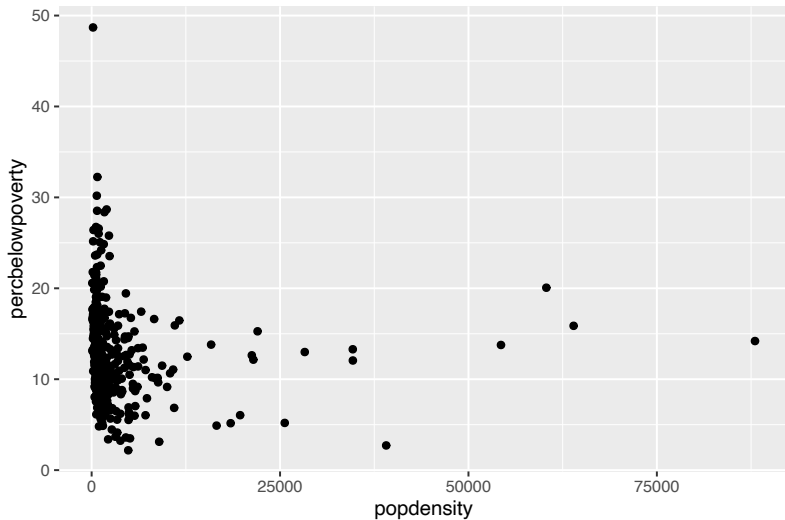
```
1 > p <- ggplot(data = midwest,  
2           mapping = aes(x = popdensity,  
3                         y = percbelowpoverty))
```

- Other aesthetic mapping parameters: color, shape, size, etc.

Adding a geom Layer

```
1 > ggplot(data = midwest,  
2         mapping = aes(x = popdensity,  
3                       y = percbelowpoverty)) +  
4     geom_point()
```

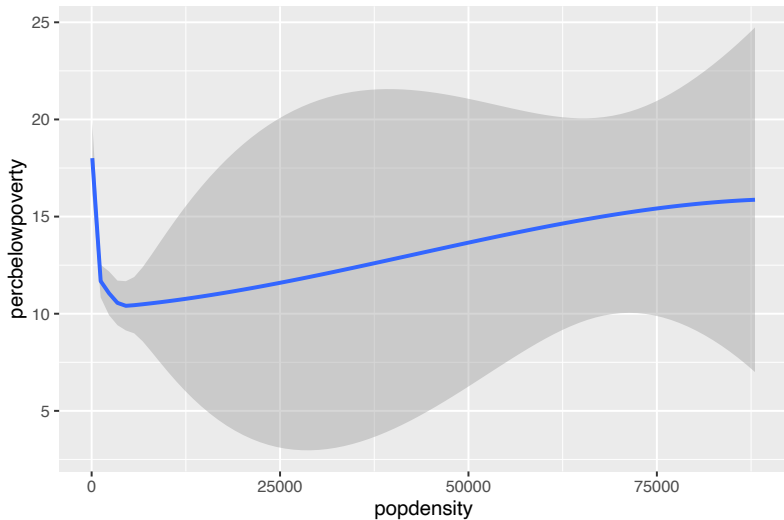
Adding a geom Layer



Trying a New geom

```
1 > ggplot(data = midwest,  
2         mapping = aes(x = popdensity,  
3                       y = percbelowpoverty)) +  
4     geom_smooth()
```

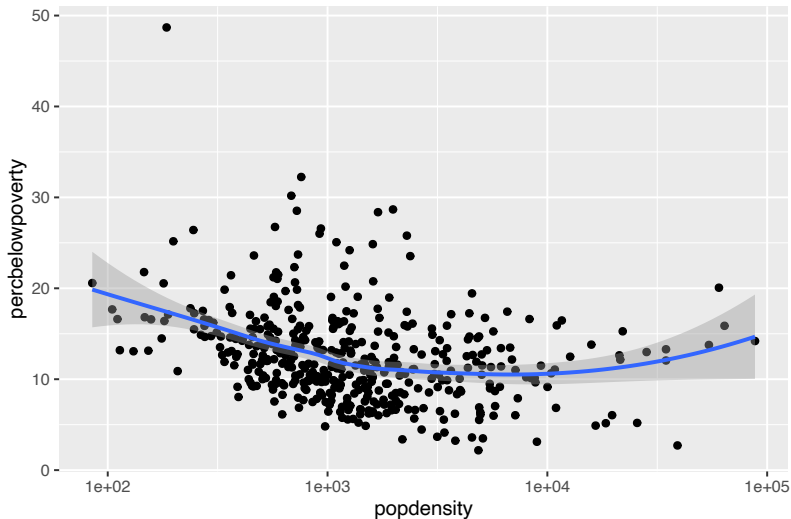
Trying a New geom



Layering geom is Additive!

```
1 > ggplot(data = midwest,  
2         mapping = aes(x = popdensity,  
3                       y = percbelowpoverty)) +  
4   geom_point() +  
5   geom_smooth() +  
6   scale_x_log10() # Log scale for the skewed x variable
```

Layering geom is Additive!

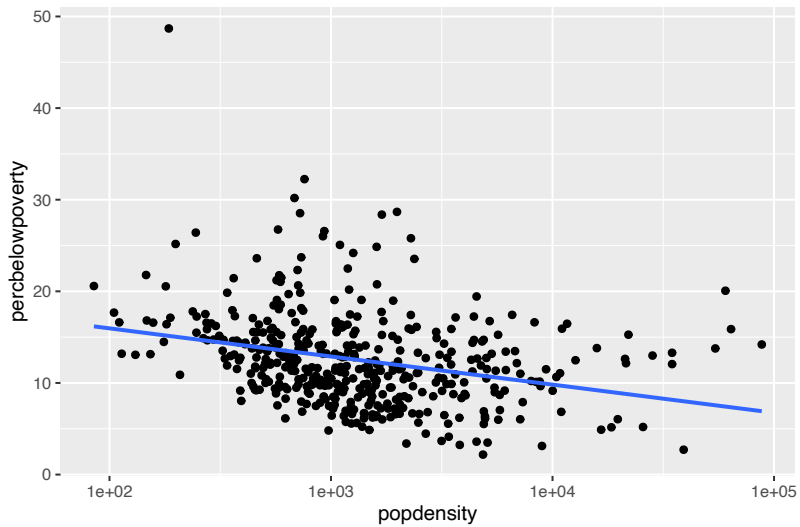


Geoms are Functions

- Geoms can take input arguments:

```
1 > ggplot(data = midwest,  
2         mapping = aes(x = popdensity,  
3                       y = percbelowpoverty)) +  
4     geom_point() +  
5     geom_smooth(method = "lm", se = F) +  
6     scale_x_log10()
```

↪ Tells `geom_smooth()` to do a linear fit with no error region.

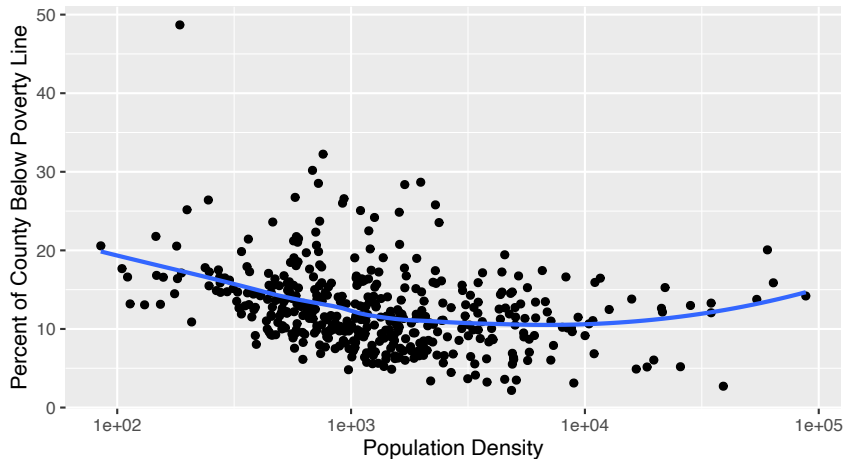


Adding Informative Labels

```
1 > ggplot(data = midwest,  
2         mapping = aes(x = popdensity,  
3                       y = percbelowpoverty)) +  
4   geom_point() +  
5   geom_smooth(method = "loess", se = F) +  
6   scale_x_log10() +  
7   labs(x = "Population Density",  
8        y = "Percent of County Below Poverty Line",  
9        title = "Poverty and Population Density",  
10       subtitle = "Among Counties in the Midwest",  
11       caption = "Source: US Census, 2000")
```

Poverty and Population Density

Among Counties in the Midwest

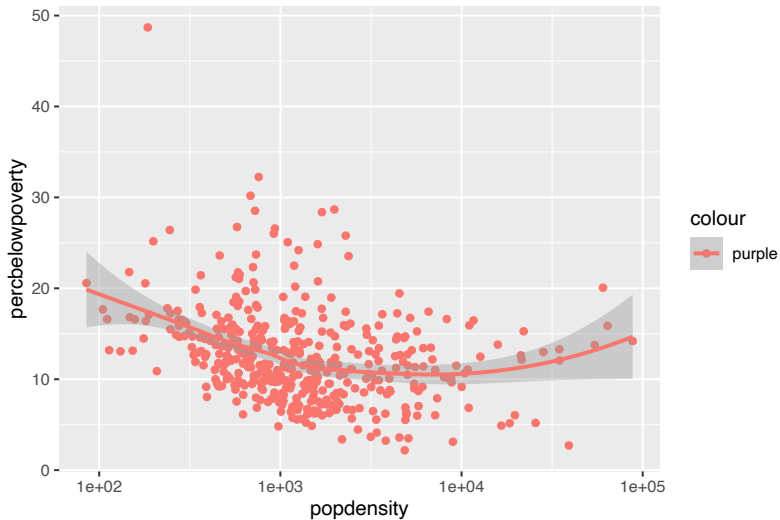


Source: US Census, 2000

Mapping vs. Setting Aesthetics

```
1 > ggplot(data = midwest,  
2         mapping = aes(x = popdensity,  
3                       y = percbelowpoverty,  
4                       color = "purple")) +  
5     geom_point() +  
6     geom_smooth() +  
7     scale_x_log10()
```

Wait, What Happened?



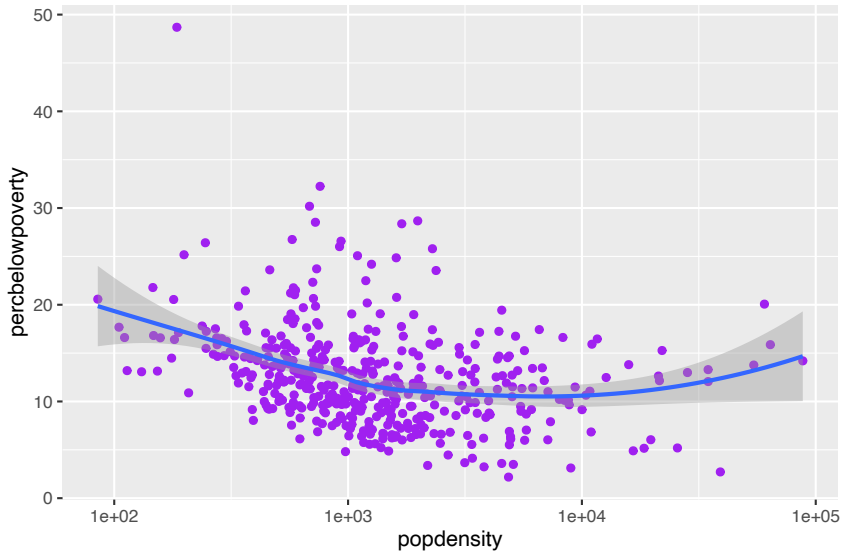
Wait, What Happened?

- Mapping always refers to variables!
 ↪ If passed a value other than a variable name, ggplot will implicitly create a variable with that value (in this case “purple” that is constant.)

Setting Aesthetics

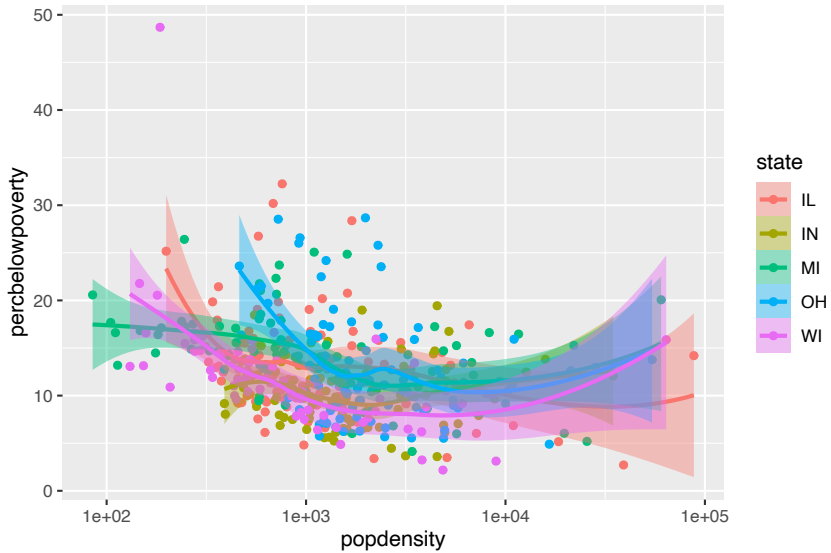
- Set the color outside the mapping (i.e., `aes()`) format:

```
1 > ggplot(data = midwest,  
2         mapping = aes(x = popdensity,  
3                       y = percbelowpoverty)) +  
4     geom_point(color = "purple") +  
5     geom_smooth() +  
6     scale_x_log10()
```



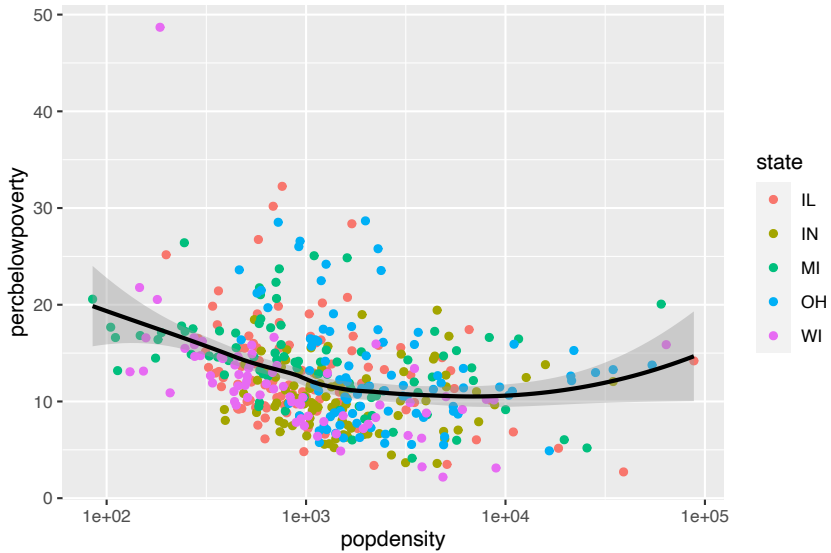
Mapping More Aesthetics

```
1 > ggplot(data = midwest,  
2         mapping = aes(x = popdensity,  
3                       y = percbelowpoverty,  
4                       color = state,  
5                       fill = state)) +  
6   geom_point() +  
7   geom_smooth() +  
8   scale_x_log10()
```



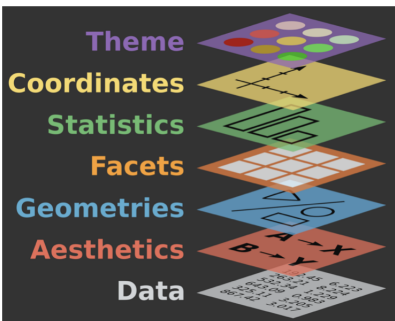
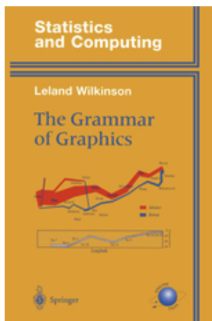
Mapping Can be Done on a Per Geom Basis

```
1 > ggplot(data = midwest,  
2         mapping = aes(x = popdensity,  
3                       y = percbelowpoverty)) +  
4   geom_point(mapping = aes(color = state)) +  
5   geom_smooth(color = "black") +  
6   scale_x_log10()
```



ggplot2: The Grammar of Graphics

“...the grammar tells us that a statistical graphic is a mapping from data to aesthetic attributes (colour, shape, size) of geometric objects (points, lines, bars). The plot may also contain statistical transformations of the data and is drawn on a specific coordinate system” – Leland Wilkinson



2/ Histograms, Boxplots, and Grouped Data

Midwest Data Redux

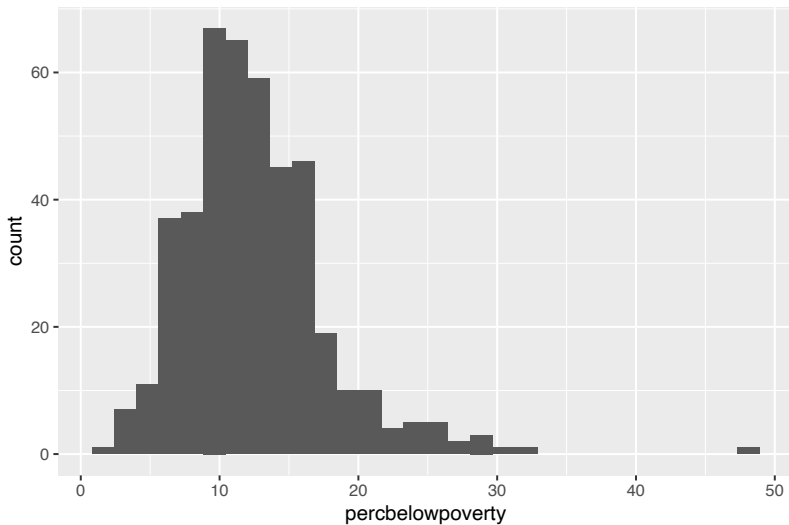
```
1 > ggplot2::midwest ## If needed: install.packages("ggplot2")
2
3 # A tibble: 437 × 28
4   PID county state area poptotal popdensity popwhite popblack popamerindian popasian popother
5   <int> <chr> <chr> <dbl> <int> <dbl> <int> <int> <int> <int> <int>
6 1 561 ADAMS IL 0.052 66090 1271. 63917 1702 98 249 124
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20 # percchildbelowpovert <dbl>, percadultpoverty <dbl>, percelderlypoverty <dbl>, inmetro <int>,
21 # category <chr>
22 # i Use `print(n = ...)` to see more rows
```

Histograms

- **Histograms** show where there are more or fewer observations of a numeric variable.

```
1 > require(ggplot2)
2 > ggplot(data = midwest,
3         mapping = aes(x = percbelowpoverty)) +
4     geom_histogram()
```

- Split up range of variable into bins, count how many are in each bin.
- y aesthetic calculated automatically.



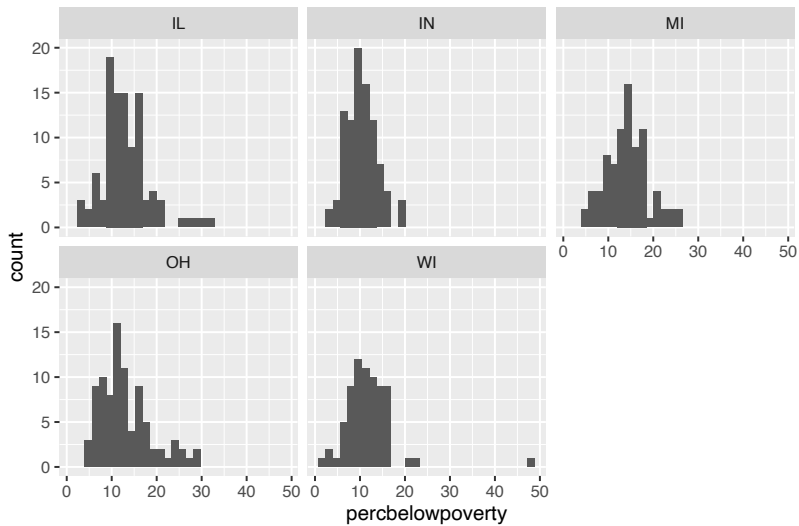
Take away: a majority of the counties have around 10% of their population living below the poverty line.

Creating Small Multiples with Facets

- Small multiples: a series of similar graphs with the same scale/axes to help with comparing different partitions of a dataset

```
1 > ggplot(data = midwest,  
2         mapping = aes(x = percbelowpoverty)) +  
3     geom_histogram() +  
4     facet_wrap(~ state)
```

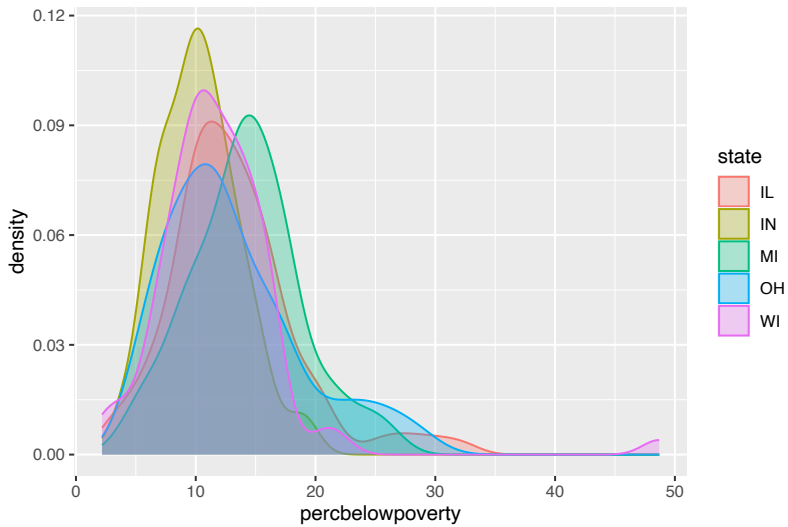
- The variable used for faceting/splitting must be placed on the right-hand side of the tilde (~) operator
 - We'll see more of the ~ variable syntax (called a formula) throughout the semester!



Density as Alternative to Histogram

- A kernel density plot is a smoothed version of a histogram and slightly easier to overlay.

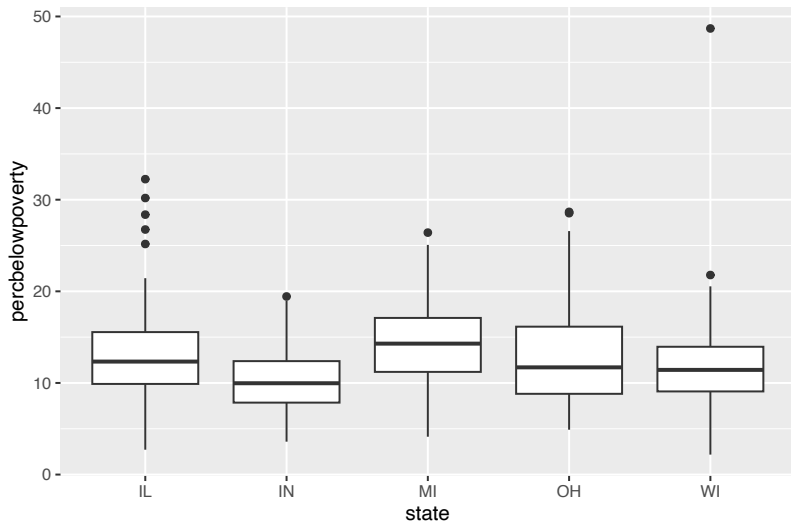
```
1 > ggplot(data = midwest,  
2         mapping = aes(x = percbelowpoverty,  
3                       fill = state, color = state)) +  
4   geom_density(alpha = 0.3)
```



Boxplots

- Boxplots are another way to compare distributions across discrete groups.

```
1 > ggplot(data = midwest,  
2         mapping = aes(x = state,  
3                       y = percbelowpoverty)) +  
4     geom_boxplot()
```



More on Boxplots

- “Box” represents middle 50% of the data.
 - 25% of the data above the box, 25% below
 - Height of the box is called the inter quartile range (IQR)
- Horizontal line in the box is the median
 - 50% of the data above the median, 50% below
- “Whiskers” represents either:
 - $1.5 \times \text{IQR}$ or max/min of the data, whichever is smaller
 - Data points beyond whiskers are outliers

Visualizing grouped data accurately is key to conveying the right insights and avoiding misleading interpretation.

Gapminder Data

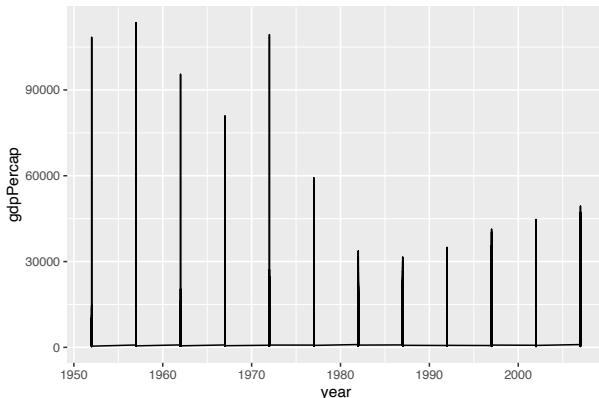
```
1 > require(gapminder)
2 > dplyr::glimpse(gapminder)
3
4 Rows: 1,704
5 Columns: 6
6 $ country   <fct> "Afghanistan", "Afghanistan", "Afghanistan", "Afghanistan", "Afgha...
7 $ continent <fct> Asia, Asia, Asia, Asia, Asia, Asia, Asia, Asia, Asia, Asia, Asia, ...
8 $ year      <int> 1952, 1957, 1962, 1967, 1972, 1977, 1982, 1987, 1992, 1997, 2002, ...
9 $ lifeExp   <dbl> 28.801, 30.332, 31.997, 34.020, 36.088, 38.438, 39.854, 40.822, 41...
10 $ pop       <int> 8425333, 9240934, 10267083, 11537966, 13079460, 14880372, 12881816...
11 $ gdpPercap <dbl> 779.4453, 820.8530, 853.1007, 836.1971, 739.9811, 786.1134, 978.01...
```

- Let's say we are interested in learning about the trend in income in different countries.
 - ↪ Use `geom_line()` to create a line plot.

Let's plot the trend in income.. wait what?

```
1 > ggplot(data = gapminder,  
2         mapping = aes(x = year,  
3                       y = gdpPercap)) +  
4     geom_line() # Didn't specify how to group the lines..
```

Let's plot the trend in income.. wait what?

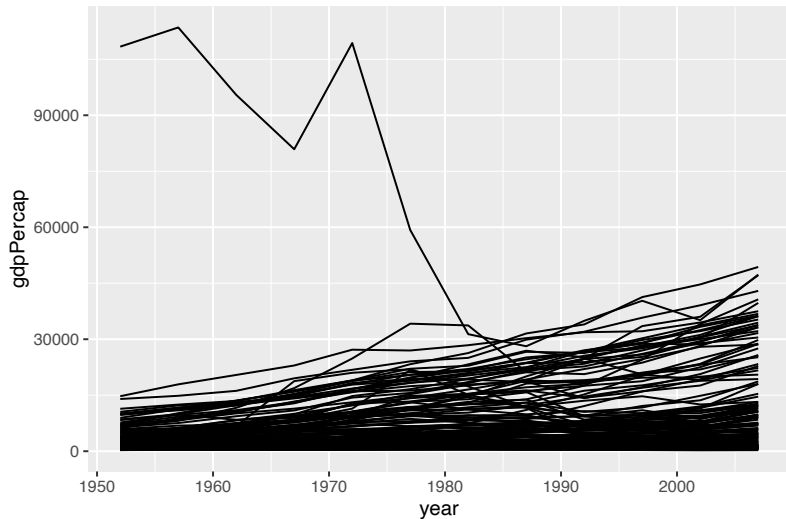


- `geom_line()` connected points from different countries in the same year.
 - What we wanted was to connect the points from same countries in different years (i.e., time trends)!

Tell `geom_line()` how to group the lines

```
1 > ggplot(data = gapminder,  
2         mapping = aes(x = year,  
3                       y = gdpPercap)) +  
4   geom_line(mapping = aes(group = country))
```

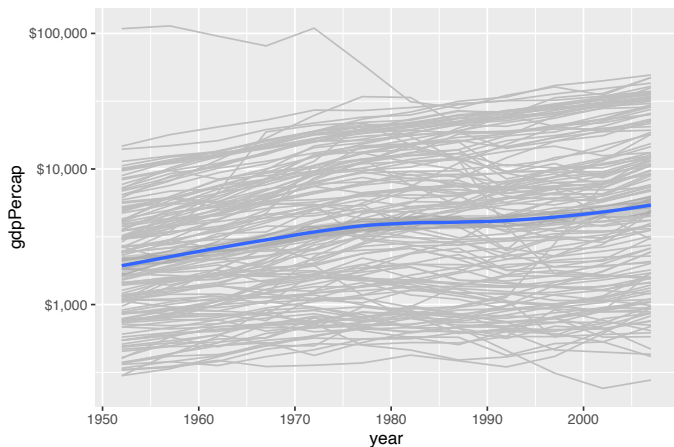
Tell `geom_line()` how to group the lines



Adjusting scales and adding a trendline

```
1 > ggplot(data = gapminder,  
2         mapping = aes(x = year,  
3                       y = gdpPercap)) +  
4   geom_line(mapping = aes(group = country), color = "gray") +  
5   geom_smooth(method = "loess") +  
6   scale_y_log10(labels = scales::dollar)
```

Adjusting scales and adding a trendline



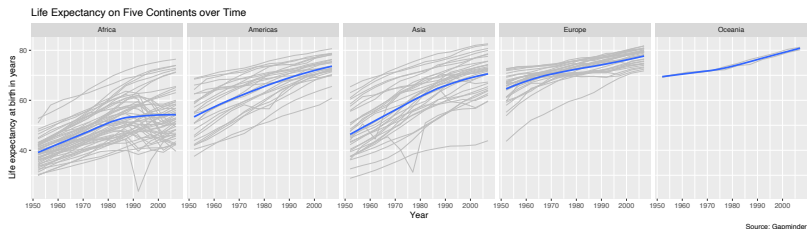
Q: What do we observe? Why do you think this is the case? (innovation? capital accumulation?)

In-Class Exercises: Background

- We want to find out the answer to the following question:
 - Q: how does life expectancy on five continents (e.g., Asia, Europe, Americas) change over time?
- The gapminder dataset contains some information about life expectancy across different continent, thus we could visualize the data to answer the question.
 - Some relevant variables:
 - `country`: name of the country
 - `continent`: name of the country's continent
 - `year`: year of the measurement, ranging from 1952 to 2007 in 5-year increments
 - `lifeExp`: life expectancy at birth, in years

In-Class Exercises: Task

- Recreate the following plot:



- The plot shows each country's life expectancy trajectory over time, broken out by continent with smoothed average lines overlayed for each continent.
 - To get started, we'll give you a few clues about what was done to create the plot in the following slide.

In-Class Exercises: Instructions

- The lines for each country use the color "gray"
- The width of the smoothed line is 1.1 (`linewidth = 1.1`) and the method used is the loess smoother.
- Turn off the standard errors.
- Make sure that the facets are all on one row. Look at the `facet_wrap` documentation if need help with this.
- Make sure that the labels are correctly specified.

Q: What does the visualization reveal? Explain what you observe verbally.

What Comes Next?

- Today: a table became a graph, layer by layer; the same data compared across distributions and groups.
- Next week: **data wrangling** with `dplyr`. Choose rows and columns, create variables, summarize groups, and draw barplots from the summaries.
- Before then: read the R basics reference slides and AAG Ch. 4–6, and finish swirl lessons 1–3. Assignment 1 is out today.
- **E1 (today's exercise): submit on eeclass by Wednesday, Sep 16, 23:59.** Exercises are normally due the same day; the first one gets two extra days.